

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 6 Summer 2026 (Regular)

Course : Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTEEOE605B - POWER PLANT ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

- 1 A thermal power station operating continuously at rated load for the entire year will have:
A. Low plant factor
B. Plant factor equal to unity
C. Zero load factor
D. Negative diversity factor
- 2 Which source of power generation is most suitable for remote areas where fuel transportation is difficult?
A. Coal-based plant
B. Diesel power plant
C. Solar power plant
D. Steam power plant
- 3 In economic operation of power plants, the load duration curve is mainly used to:
A. Measure turbine speed
B. Estimate energy requirements
C. Calculate transmission losses only
D. Determine insulation resistance
- 4 The moderator used in a nuclear reactor primarily functions to:
A. Absorb all neutrons
B. Slow down fast neutrons
C. Increase steam pressure
D. Cool the turbine blades
- 5 In a thermal power plant, the economizer is used to:
A. Increase turbine speed
B. Heat feed water using flue gases
C. Condense steam
D. Store coal

- 6 Surge tanks are generally installed in hydroelectric power plants to:
 - A. Store coal
 - B. Control water hammer effect
 - C. Increase generator voltage
 - D. Reduce transformer losses
 - 7 Diesel power plants are most suitable for:
 - A. Base load operation only
 - B. Peak load and standby operation
 - C. Nuclear applications
 - D. Continuous hydro operation
 - 8 In gas turbine plants, regeneration improves efficiency by:
 - A. Cooling turbine blades
 - B. Utilizing exhaust heat to preheat air
 - C. Increasing fuel consumption
 - D. Increasing condenser pressure
 - 9 Wind power generation is most economical in areas having:
 - A. Low wind velocity
 - B. Uniform moderate wind speed
 - C. No atmospheric pressure
 - D. High humidity only
 - 10 Which renewable source provides continuous power independent of weather conditions?
 - A. Solar energy
 - B. Wind energy
 - C. Geothermal energy
 - D. Tidal energy only
 - 11 DC supply in power plants is mainly required for:
 - A. Boiler operation only
 - B. Emergency control and protection circuits
 - C. Coal transportation
 - D. Cooling tower operation
 - 12 The tariff charged to industrial consumers generally includes:
 - A. Fixed charge only
 - B. Energy charge only
 - C. Fixed and maximum demand charges
 - D. No energy charges
- Q2. Solve the following.
- A) Draw the schematic layout of any two of the following:
 - i. Thermal Power Plant
 - ii. Hydro Electric Power Plant
 - iii. Nuclear Power Plant
 - B) A generating station has: Plant capacity = 120 kW, Capacity factor = 0.65, Demand factor = 0.78 and Annual operation = 8760 hours. (Assume Load Factor = Capacity Factor) Calculate:
 - i. Average Load
 - ii. Maximum Demand
 - iii. Connected Load
 - iv. Annual Energy Generated

- Q3. Solve the following.
- A) Describe the Classification of various Turbines used in Hydro-electric power plant. 6
 - B) Apply the concept of hydraulic head and discharge to calculate the power output of a hydroelectric power plant when the head is 140 m and flow rate is 45 m³/s. Assume overall efficiency is 82%. 6
- Q4. Solve Any Two of the following.
- A) Describe the construction and working of a nuclear reactor. Analyze the safety measures in nuclear power plants. 6
 - B) Compare diesel power plants and gas turbine power plants based on operation and applications. 6
 - C) Analyze the importance of regeneration and intercooling in gas turbine plants. 6
- Q5. Solve Any Two of the following.
- A) Explain the principle, construction, and working of an MHD power generation system. 6
 - B) Analyze the Working Operations of Single Basin and Double Basin Tidal Power Generation Plants 6
 - C) Evaluate the role of biomass and geothermal energy in sustainable power generation. 6
- Q6. Solve Any Two of the following.
- A) Explain different types of electricity tariffs and their practical applications. 6
 - B) Explain the working and importance of switchyards in power systems. 6
 - C) Discuss the importance of power plant instrumentation and control systems. 6

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